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- 2 classes, 0 - is not a school bus, 1 is a school Bus
- split the 30×30 image into a 10×10 tiles to pass each image one at a time to the CNN
- A $10 \times 10 \times 3$ ~~image~~ with RGB channels, (10×10 with 3 channels)
- the output dimension should be a 1×2 vector
- the postprocessing should include running the softmax function to determine which class has the highest probability and using that to determine and set labels to each image

2) $X = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix}$

$K = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

a) Y will be a 3×3 matrix

$a_{11} = 1+0+1+0+1+0+1+0+0 = 4$

$a_{12} = 0+0+0+0+1+0+1+0+1 = 3$

$a_{13} = 1+0+1+0+0+0+0+0+0 = 2$

$a_{31} = 1+0+0+0+0+0+1+0+0 = 2$

$a_{32} = 1+0+1+0+1+0+0+0+1 = 4$

$a_{33} = 0+0+0+0+1+0+0+0+0 = 1$

$a_{21} = 0+0+1+0+1+0+0+0+1 = 3$

$a_{22} = 1+0+0+0+0+0+0+0+1 = 2$

$a_{23} = 1+0+0+0+0+1+0+1+0+1 = 4$

b) $Y = \begin{bmatrix} 4 & 3 & 2 \\ 3 & 2 & 4 \\ 2 & 4 & 1 \end{bmatrix}$

$P_{11} = (4+3+3+2)/4 = 3$ $P_{12} = (3+2+2+4)/4 = 11/4$

$P_{21} = (3+2+2+4)/4 = 11/4$ $P_{22} = (2+4+4+1)/4 = 11/4$

c) P is a 2×2 matrix

d) $P = \begin{bmatrix} 3 & 11/4 \\ 11/4 & 11/4 \end{bmatrix}$

3) $e^2 = 7.389$, $e^0 = 1$, $e^{-1} = 0.368$

$\sum e^z = 9.757$

a) $\text{softmax}(z) = \left[\frac{7.389}{9.757}, \frac{1}{9.757}, \frac{0.368}{9.757} \right] = [0.744, 0.114, 0.042]$

b) index 1 is the predicted class with the highest softmax